

HK IMO Training 2019-2020
Problem Set 11
For 11 January 2020



Problem 41. Let ABC be an acute-angled triangle with $AB > AC$, circumcentre O and orthocentre H . Denote by D , E , and F the feet of the perpendiculars from A , B , and C to BC , CA , and AB , respectively. If X is the reflection of A about EF , prove that O , X , D , and H are concyclic.

Problem 42. Let ABC be an acute-angled triangle with $AB = AC$. Suppose that D is the foot of the perpendicular from A to BC , M is the midpoint of AD , and E is foot of the perpendicular from D to CM . Prove that $\angle AEB = 90^\circ$.

Problem 43. Let ABC be an acute-angled triangle with $AB > AC$ and orthocentre H . Suppose that E and F are the feet of the perpendiculars from B and C to CA and AB , respectively, M and N are the midpoints of BC and EF , respectively, and M' is the reflection of M about BE . Prove that $\angle M'NH = 90^\circ$.

Problem 44. Let ABC be an acute-angled triangle with $AB > AC$. Suppose that D is the foot of the perpendicular from A to BC , E and F are the midpoints of CA and AB , respectively, and X is point inside $\triangle ABC$ such that $\angle CXA = \angle FXD = 90^\circ$. If X and B do not lie on the same side with respect to CF , prove that EF bisects $\angle CFX$.